KERALA SSLC | MATHEMATICS

# ARITHMETIC SEQUENCES

PYQ • 2021 - 2026

**1.** 2026 **1 MARK**

In which of the arithmetic sequences given below, 37 is a term?

(a) 4, 9, 14, … (b) 7, 12, 17, … (c) 8, 12, 16, … (d) 7, 11, 15, …

**2.** 2026 **1 MARK**

Which of the following is the algebraic form of the arithmetic sequence 6, 10, 14, … ?

(a) 4n + 1 (b) 6n + 4 (c) 5n + 1 (d) 4n + 2

**3.** 2026 **2 MARKS**

The 7th term of an arithmetic sequence is 100 and the 11th term is 140.

(i) What is the fifteenth term of this sequence? **(1)**

(ii) What is the third term of this sequence? **(1)**

**4.** 2026 **4 MARKS**

(A) The sum of the first 3 terms of an arithmetic sequence is 30 and the sum of the first 7 terms is 140.

(i) What is the second term of this sequence? **(1)**

(ii) What is the fourth term of this sequence? **(1)**

(iii) Write the first three terms of this sequence. **(2)**

OR

(B) The sum of the first and 25th terms of an arithmetic sequence is 70.

(i) What is the sum of the 4th and 22nd terms of this sequence? **(1)**

(ii) What is the 13th term of this sequence? **(1)**

(iii) Find the sum of the first 25 terms of this sequence. **(2)**

**5.** 2026 **4 MARKS**

(i) Write the sequence of natural numbers which leaves a remainder 1 on division by 4. **(1)**

(ii) Is this an arithmetic sequence? Why? **(1)**

(iii) Write the algebraic form of the arithmetic sequence got by adding 2 to each term of the above sequence. **(2)**

**6.** 2026 **4 MARKS**

(A) The sum of the first n terms of an arithmetic sequence is $n^2 + 6n$.

(i) What is the sum of the first 7 terms? **(1)**

(ii) How many terms of this sequence, starting from the first, must be added to get 315? **(3)**

OR

(B) 5, 11, 17, … is an arithmetic sequence.

(i) Write the algebraic form of this sequence. **(1)**

(ii) Calculate the sum of the first 15 terms of this sequence. **(2)**

(iii) Write the sum of the first n terms of this sequence. **(1)**

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**7.** 2025 **2 MARKS**

3, 8, 13, … is an arithmetic sequence.

(a) What is the common difference?

(b) What is its 11th term?

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**8.** 2025 **3 MARKS**

Consider the arithmetic sequence 2, 8, 14, …

(a) What is the remainder obtained when the terms of this sequence are divided by 6?

(b) Is 176 a term of this sequence? Why?

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**9.** 2025 **4 MARKS**

Find the sum.

(a) 1 + 2 + 3 + … + 20

(b) 5 + 10 + 15 + … + 100

(c) 8 + 13 + 18 + … + 103

(d) 4 + 9 + 14 + … + 99

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**10.** 2025 **5 MARKS**

The 6th term of an arithmetic sequence is 27 and the 16th term is 67.

(a) Find the common difference of this sequence.

(b) Find its first term.

(c) Write its algebraic form.

(d) Find the sum of the first 31 terms of this sequence.

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**11.** 2024 **2 MARKS**

The sequence of perimeters of squares of sides 1 centimetre, 2 centimetres, 3 centimetres and so on form an arithmetic sequence.

(a) Write the sequence.

(b) What is the common difference?

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**12.** 2024 **4 MARKS**

Consider the arithmetic sequence 8, 14, 20, …

(a) Is 25 a term of this sequence?

(b) Check if 144 is a term of this sequence.

(c) Prove that there are no perfect squares in this sequence.

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**13.** 2024 **5 MARKS**

The third term of an arithmetic sequence is 26 and its eighth term is 61.

(a) Find the common difference of the sequence.

(b) What is its first term?

(c) Write the algebraic form of the sequence.

(d) Find the sum of the first 15 terms of the sequence.

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**14.** 2023 **2 MARKS**

7, 13, 19, … is an arithmetic sequence.

(a) What is its common difference?

(b) Find its 11th term.

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**15.** 2023 **3 MARKS**

The 6th term of an arithmetic sequence is 46. Its common difference is 8.

(a) What is its 16th term?

(b) Find its 21st term.

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**16.** 2023 **4 MARKS**

The sum of the first 7 terms of an arithmetic sequence is 140. The sum of the first 11 terms of the same arithmetic sequence is 440.

(a) What is the 4th term of this arithmetic sequence?

(b) Find its 6th term.

(c) What is the common difference?

(d) Find the first term of this sequence.

**17.** 2023 **5 MARKS**

(a) Find the sum of the first 20 natural numbers.

(b) Write the algebraic expression of the arithmetic sequence 5, 9, 13, …

(c) Find the sum of the first 20 terms of the arithmetic sequence 5, 9, 13, …

**18.** 2022 **1 MARK**

What is the common difference of the arithmetic sequence 3, 7, 11, … ?

**19.** 2022 **2 MARKS**

5, 8, 11, … is an arithmetic sequence.

(a) What is its 20th term? **(1)**

(b) What is the algebraic expression for this sequence? **(1)**

**20.** 2022 **2 MARKS**

The algebraic expression for the sum of n terms of an arithmetic sequence is $n^2 + n$.

(a) Find the first term of this arithmetic sequence. **(1)**

(b) Find the sum of the first 10 terms of this arithmetic sequence. **(1)**

**21.** 2022 **8 MARKS**

6, 10, 14, … is an arithmetic sequence.

(a) Find the sum of the first 15 terms of this arithmetic sequence. **(4)**

(b) What is the difference between the first term and the 16th term? **(2)**

(c) Find the difference between the sum of the first 15 terms and the sum of the next 15 terms. **(2)**

**22.** 2021 **1 MARK**

The arithmetic sequence with common difference 2 is:

(7, 10, 13, … ; 7, 5, 3, … ; 7, 9, 11, … ; 2, 5, 8, …)

**23.** 2021 **2 MARKS**

Write the first term and common difference of the arithmetic sequence 3n + 2.

**24.** 2021 **3 MARKS**

(a) What is the tenth term of the arithmetic sequence a + 1, a + 2, a + 3, … ? **(1)**

(b) What is its common difference? **(1)**

(c) Write the algebraic form of the above sequence. **(1)**

**25.** 2021 **4 MARKS**

(a) Write the 20th term of the arithmetic sequence 5, 10, 15, … **(1)**

(b) Find the sum of the first 20 terms of the arithmetic sequence 5, 10, 15, … **(2)**

(c) What is the sum of the first 20 terms of the arithmetic sequence 4, 9, 14, … ? **(1)**

**26.** 2021 **5 MARKS**

1

2 3

4 5 6

7 8 9 10

…

(a) Write the fifth line of the pattern. **(1)**

(b) How many numbers are there in the tenth line? **(1)**

(c) How many numbers are there in the first ten lines altogether? **(2)**

(d) What is the first number in the eleventh line? **(1)**

**27.** 2021 **5 MARKS**

(a) What is the remainder on dividing the terms of the arithmetic sequence 100, 109, 118, … by 9? **(1)**

(b) Write the sequence of three digit numbers which are multiples of 9. **(2)**

(c) What is the position of 999 in the arithmetic sequence of three digit numbers which are multiples of 9? **(2)**